

SANDANSKI, Bulgaria

WMO: 157120

Lat: 41.550N

Lon: 23.267E

Elev: 679

StdP: 14.34

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 20.1	(c) 24.1	(d) 11.4	(e) 10.1	(f) 25.0	(g) 15.2	(h) 12.2	(i) 28.9	(j) 27.8	(k) 37.7	(l) 23.2	(m) 42.1	(n) 4.3	(o) 320	(p) 0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	97.4	71.3	94.4	70.5	91.8	69.6	73.3	92.9	71.9	90.8	70.7	88.7	4.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	101.4	81.3	65.6	96.9	80.1	64.4	92.9	79.2	37.3	93.0	36.0	91.2	34.9	88.9	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.4	(b) 18.3	(c) 14.6	(e) 15.1	(f) 102.1	(g) 5.1	(h) 3.5	(i) 11.5	(j) 104.6	(k) 8.5	(l) 106.7	(m) 5.7	(n) 108.6	(o) 2.0	(p) 111.1		
(a) 22.4	(b) 18.3	(c) 14.6	(e) 14.1	(f) 77.0	(g) 4.9	(h) 2.8	(i) 10.5	(j) 79.1	(k) 7.7	(l) 80.7	(m) 4.9	(n) 82.3	(o) 1.3	(p) 84.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.8	37.8	42.8	49.8	57.8	66.2	73.8	78.8	79.0	70.3	59.7	49.2	39.9		
	DBStd	15.65	7.51	7.50	6.76	5.90	5.59	5.84	4.84	4.76	6.08	6.69	7.11	7.31		
	HDD50	1121	380	215	86	11	0	0	0	0	0	8	98	323		
	HDD65	3674	842	620	470	223	55	5	0	0	20	185	475	779		
	CDD50	4352	3	15	81	245	503	713	893	898	608	310	73	10		
	CDD65	1431	0	0	0	7	93	268	428	434	179	22	0	0		
	CDH74	14721	0	0	3	87	811	2714	4746	4658	1541	160	1	0		
	CDH80	6764	0	0	0	10	198	1186	2395	2369	584	22	0	0		
(14) Wind	WSAvg	4.7	4.2	5.5	6.0	5.2	4.7	4.9	5.1	4.5	4.5	3.9	3.9	4.3		
(15) Precipitation	PrecAvg	21.5	1.6	1.5	1.7	2.0	2.1	1.9	1.5	1.3	1.7	2.0	1.7	2.4		
	PrecMax	32.0	4.3	5.0	3.7	4.6	4.5	4.4	4.7	3.5	4.1	5.6	3.8	5.0		
	PrecMin	10.7	0.1	0.2	0.3	0.7	1.1	0.4	0.3	0.1	0.2	0.1	0.1	0.0		
	PrecStd	4.8	1.0	1.1	0.9	1.0	0.9	1.0	1.0	1.0	1.2	1.2	0.8	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.0	66.0	73.5	81.0	88.7	97.6	102.0	100.9	94.8	83.0	70.6	62.5		
		MCWB	49.6	53.0	56.0	61.7	67.3	71.7	71.9	72.2	68.4	65.0	59.3	55.6		
	2%	DB	55.0	61.5	69.0	76.2	84.9	93.6	97.5	97.5	90.3	78.4	66.2	57.3		
		MCWB	48.2	50.5	53.8	59.4	65.9	71.0	71.3	71.6	67.1	62.9	57.8	51.8		
	5%	DB	51.4	57.7	65.4	72.8	81.9	90.3	94.4	94.4	86.7	74.7	62.9	53.0		
		MCWB	46.1	49.0	52.2	57.6	64.3	69.7	70.6	70.3	66.1	61.1	55.5	48.7		
	10%	DB	48.3	54.3	61.7	69.6	78.8	87.0	91.5	91.5	82.8	71.2	59.6	50.2		
		MCWB	44.2	47.3	50.3	55.7	63.2	68.3	69.7	69.5	64.9	59.9	53.7	46.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	54.5	58.2	63.1	69.7	74.2	74.9	75.4	70.4	67.2	60.8	56.7		
		MCDB	57.0	62.8	69.5	77.2	85.2	92.3	93.8	95.4	88.9	79.9	67.8	61.7		
	2%	WB	48.9	51.8	55.0	60.9	67.3	72.3	73.2	73.3	68.5	64.4	58.5	52.6		
		MCDB	53.1	59.5	65.9	73.8	81.8	90.4	93.0	93.4	85.8	75.1	64.2	56.5		
	5%	WB	46.8	49.9	53.2	58.8	65.5	70.7	71.9	71.8	67.2	62.7	56.7	49.7		
		MCDB	50.3	56.2	63.2	70.2	79.3	88.0	90.8	90.9	82.9	71.7	61.9	52.3		
	10%	WB	44.5	48.0	51.3	57.0	63.9	69.2	70.7	70.4	65.8	60.9	54.6	47.2		
		MCDB	48.0	53.8	60.1	67.2	76.7	85.0	88.5	88.5	79.7	69.0	58.9	49.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	15.4	17.6	18.9	20.3	22.0	23.1	23.2	20.5	17.8	14.6	12.0		
		MCDBR	16.7	20.6	23.6	24.5	25.1	26.5	27.1	26.7	25.7	22.9	18.5	14.7		
		MCWBR	10.9	11.9	12.1	11.6	10.7	10.1	10.0	9.5	9.9	10.7	10.6	9.7		
		MCDBR	13.9	18.6	20.4	22.0	23.1	25.2	25.0	24.9	22.7	19.3	15.8	11.5		
	5% WB	MCWBR	9.5	11.2	11.1	10.8	10.7	10.3	9.7	9.7	9.4	9.8	9.4	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.358	0.403	0.462	0.454	0.447	0.444	0.455	0.432	0.392	0.358	0.333		
	taud		2.393	2.306	2.206	2.050	2.147	2.209	2.222	2.188	2.224	2.323	2.379	2.388		
	Ebn at Noon		258	266	266	257	262	264	264	256	253	250	244	245		
	Edh at Noon		26	33	41	51	48	45	44	44	40	32	26	24		
(44) All-Sky Solar Radiation	RadAvg		552	822	1164	1488	1806	2046	2136	1910	1421	988	647	492		
	RadStd		81	108	125	160	110	154	120	128	114	110	61	81		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	+0.85	N/A	N/A	N/A	N/A	+0.72	+0.72	-157	-279	N/A

Nomenclature: See separate page